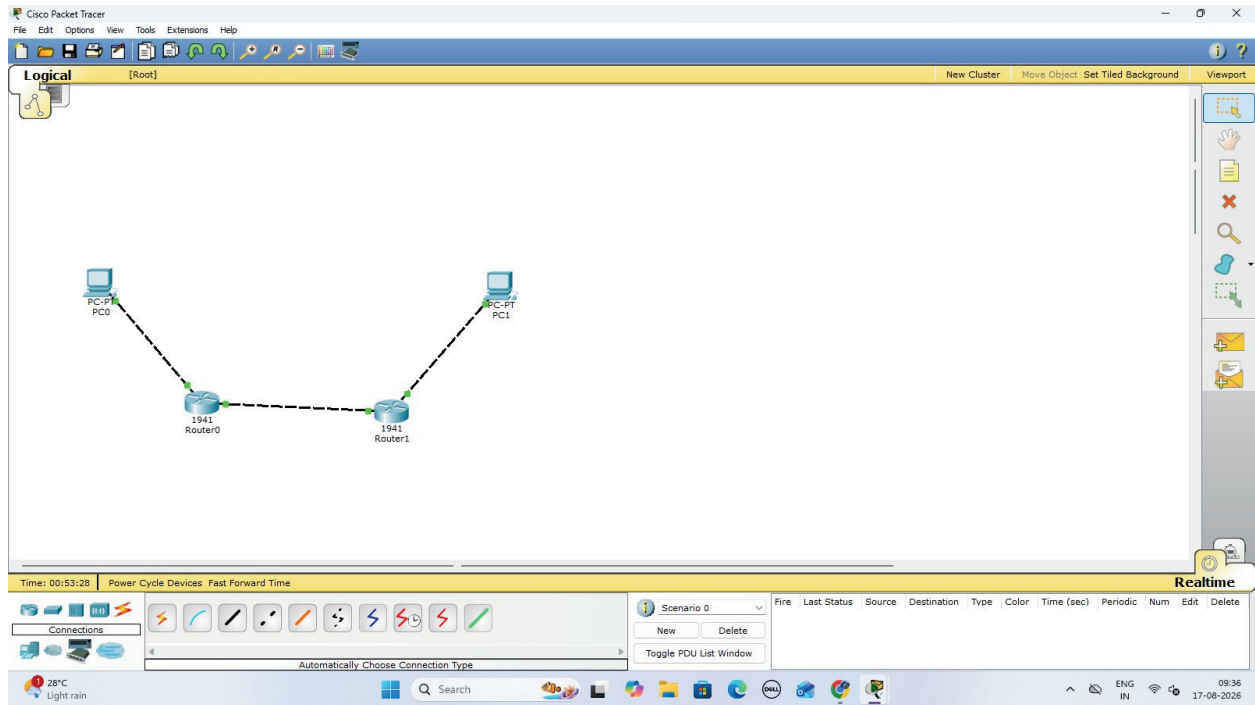


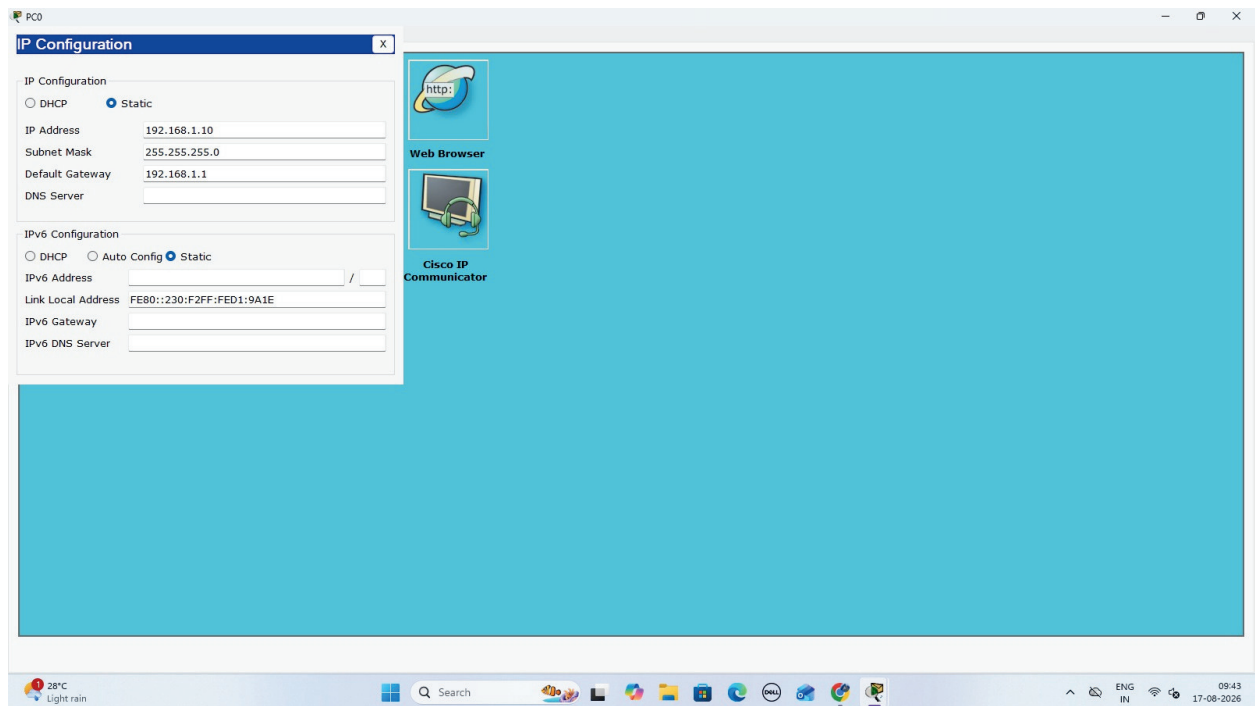
PRACTICAL NO. 6

AIM: Configure IPsec on network devices to provide secure communication and protect against unauthorized access and attacks.

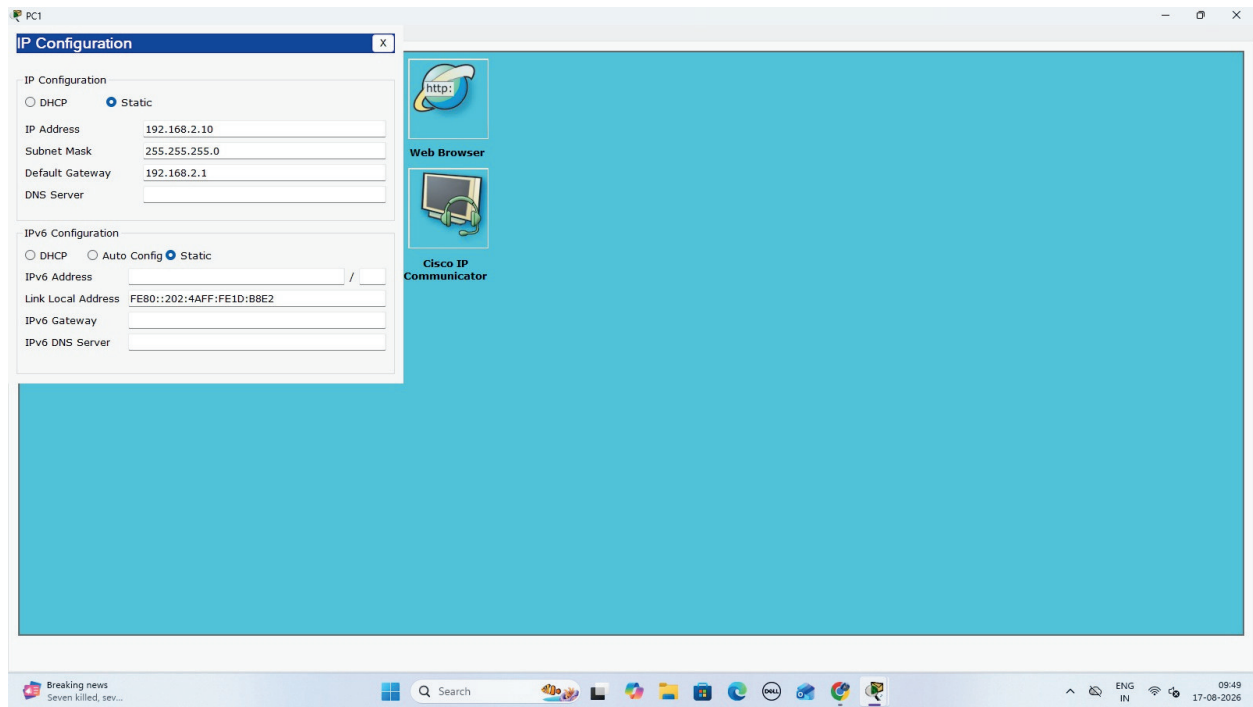
CONNECTION:



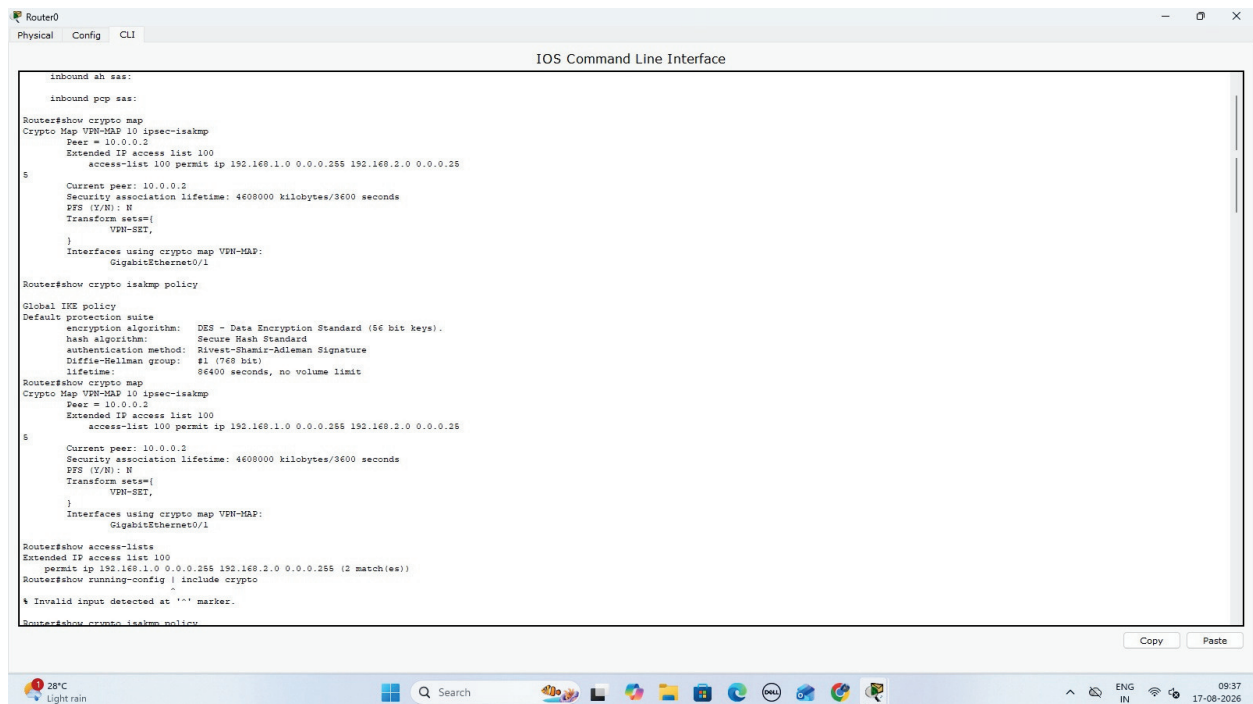
PC 0 -



PC 1 -



ROUTER 0 -



Router0

Physical Config CLI

IOS Command Line Interface

```

Router#show access-lists
Extended IP access list 100
  permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255 (2 match(es))
Router#show running-config | include crypto
* Invalid input detected at '^' marker.

Router#show crypto isakmp policy

Global IKE policy
Default protection suite
  encryption algorithm: DES - Data Encryption Standard (56 bit keys)
  hash algorithm:      Secure Hash Standard
  authentication method: Rivest-Shamir-Adleman Signature
  Diffie-Hellman group: #1 (768 bit)
  lifetime:             3600 seconds, no volume limit

Router#show crypto map
Crypto Map VPN-MAP 10 ipsec-isakmp
  peer = 10.0.0.2
  Extended IP access list 100
  access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.25
5
  Current peer: 10.0.0.2
  Security association lifetime: 4608000 kilobytes/3600 seconds
  STS (VPI): 0
  Transform sets=
    VPN-SET,
  Interfaces using crypto map VPN-MAP:
    GigabitEthernet0/1

Router#show access-lists
Extended IP access list 100
  permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255 (2 match(es))
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#hash sha
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 5
Router(config-isakmp)#exit
Router(config)#crypto isakmp key cisco123 address 10.0.0.2
A pre-shared key for address mask 10.0.0.2 255.255.255.255 already exists!
Router(config)#end
Router#
*SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief

```

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Router0

Physical Config CLI

IOS Command Line Interface

```

Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  192.168.1.1    YES manual up          up
GigabitEthernet0/1  10.0.0.1       YES manual up          up
Vlan1           unassigned     YES NVRAM  administratively down down
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/30 is directly connected, GigabitEthernet0/1
L    10.0.0.1/32 is directly connected, GigabitEthernet0/1
L    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, GigabitEthernet0/0
L    192.168.1.1/32 is directly connected, GigabitEthernet0/0
S    192.168.2.0/24 (1/0) via 10.0.0.2

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/30 is directly connected, GigabitEthernet0/1
L    10.0.0.1/32 is directly connected, GigabitEthernet0/1
L    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, GigabitEthernet0/0
L    192.168.1.1/32 is directly connected, GigabitEthernet0/0
S    192.168.2.0/24 (1/0) via 10.0.0.2

Router#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status
10.0.0.2     10.0.0.1     QM_IDLE        1065         0 ACTIVE

```

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Router0

Physical Config CLI

IOS Command Line Interface

```

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C   10.0.0.0/30 is directly connected, GigabitEthernet0/1
L   10.0.0.1/32 is directly connected, GigabitEthernet0/1
C   192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C   192.168.1.0/24 is directly connected, GigabitEthernet0/0
L   192.168.1.1/32 is directly connected, GigabitEthernet0/0
S   192.168.2.0/24 (11/0) via 10.0.0.2

Router#show crypto isakmp sa
dst          src          state      conn-id slot status
10.0.0.2     10.0.0.1     QM_IDLE    1085     0 ACTIVE

IPv6 Crypto ISAKMP SA

Router#show crypto ipsec sa
interface: GigabitEthernet0/1
Crypto map tag: VPN-MAP, local addr 10.0.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.2.0/255.255.255.0/0/0)
current_peer 10.0.0.2 port 500
PERMIT, flags={origin_is_acl,}
#pkts encaps: 7, #pkts encrypt: 7, #pkts digest: 0
#pkts decaps: 6, #pkts decrypt: 6, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 1, #recv errors 0

local crypto endpt.: 10.0.0.1, remote crypto endpt.: 10.0.0.2
path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/1
current outbound spi: 0x471F1F1E(1193221918)

inbound esp sas:
spi: 0x106D7C7C(275610748)
transform: esp-aes esp-sha-hmac ,

Router#show crypto ipsec sa
interface: GigabitEthernet0/1
Crypto map tag: VPN-MAP, local addr 10.0.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.2.0/255.255.255.0/0/0)
current_peer 10.0.0.2 port 500
PERMIT, flags={origin_is_acl,}
#pkts encaps: 7, #pkts encrypt: 7, #pkts digest: 0

```

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Router0

Physical Config CLI

IOS Command Line Interface

```

protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.2.0/255.255.255.0/0/0)
current_peer 10.0.0.2 port 500
PERMIT, flags={origin_is_acl,}
#pkts encaps: 7, #pkts encrypt: 7, #pkts digest: 0
#pkts decaps: 6, #pkts decrypt: 6, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 1, #recv errors 0

local crypto endpt.: 10.0.0.1, remote crypto endpt.: 10.0.0.2
path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/1
current outbound spi: 0x471F1F1E(1193221918)

inbound esp sas:
spi: 0x106D7C7C(275610748)

Router#show crypto isakmp sa
IPv6 Crypto ISAKMP SA
dst          src          state      conn-id slot status
10.0.0.2     10.0.0.1     QM_IDLE    1085     0 ACTIVE

IPv6 Crypto ISAKMP SA

Router#show crypto ipsec sa
interface: GigabitEthernet0/1
Crypto map tag: VPN-MAP, local addr 10.0.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.2.0/255.255.255.0/0/0)
current_peer 10.0.0.2 port 500
PERMIT, flags={origin_is_acl,}
#pkts encaps: 11, #pkts encrypt: 11, #pkts digest: 0
#pkts decaps: 10, #pkts decrypt: 10, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 1, #recv errors 0

local crypto endpt.: 10.0.0.1, remote crypto endpt.: 10.0.0.2
path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/1
current outbound spi: 0x471F1F1E(1193221918)

inbound esp sas:
spi: 0x106D7C7C(275610748)
--More--

```

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ROUTER 1 -

```

Router1
Physical Config CLI

IOS Command Line Interface

ACCEPT? [yes/no]: yes
% use 'write' command to make license boot config take effect on next boot
%IOS_LICENSE_HIDE APPLICATION=IOS-LICENSE_APP: Mobile name = C1900 Next reboot
level = security9 and license = security9

Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#reload
Proceed with reload? [confirm]
System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 2010 by Cisco Systems, Inc.
Total memory size = 512 MB - On-board = 512 MB, DIMM0 = 0 MB
CISCO1941/F9 platform with 524858 bytes of main memory
Main memory is configured to 64/-1(On-board/DIMM0) bit mode with ECC disabled

Readonly ROMMON initialized
program load complete, entry point: 0x80803000, size: 0x1b340
program load complete, entry point: 0x80803000, size: 0x1b340

IOS Image Load Test

Digitally Signed Release Software
program load complete, entry point: 0x10000000, size: 0x2bb1c58
Self decompressing the image :
##### [OK]
Smart Init is enabled
smart init is sizing iomem
      TYPE      MEMORY_REQ
Onboard devices &
  buffer pools  0x01E8F000
-----
TOTAL:         0x01E8F000
Rounded IOMEM up to: 32Mb.
Using 6 percent iomem. [32Mb/512Mb]

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subject to restrictions as set forth in subparagraph
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Software clause at DFARS sec. 252.227-7013.

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```

```

Router1
Physical Config CLI

IOS Command Line Interface

170 West Tasman Drive
San Jose, California 95134-1706

Cisco IOS Software, C1900 Software (C1900-UNIVERSALK9-M), Version 15.1(4)M4, REL
EASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2012 by Cisco Systems, Inc.
Compiled Thu Jul 5-Jan12 15:41 by pt_team
Image text-base: 0x2100F918, data-base: 0x24729040

This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wai/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco C19001941/F9 (revision 1.0) with 491620K/31748K bytes of memory.
Processor board ID FTL18400RS
2 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
256K bytes of non-volatile configuration memory.
249864K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINK-5-CHANGED: Interface Vlan1, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed stat
e to up
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed stat
e to up
%LINK-5-CHANGED: Interface Vlan1, changed state to administratively down
%SYS-5-CONFIG_I: Configured from console by console

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```

```

Router1
Physical Config CLI

IOS Command Line Interface

%LINEPROTO-6-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed stat
e to up
%LINK-6-CHANGED: Interface Vlan1, changed state to administratively down
%SYS-6-CONFIG_I: Configured from console by console

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#hash sha
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 5
Router(config-isakmp)#
Router(config-isakmp)#exit
Router(config)#
%LINEPROTO-6-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed stat
e to down
%LINEPROTO-6-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed stat
e to up

Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#hash sha
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 5
Router(config-isakmp)#exit
Router(config)#crypto isakmp key cisco123 address 10.0.0.1
Router(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
Router(config)#access-list 100 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0
.255
Router(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Router(config-crypto-map)#set peer 10.0.0.1
Router(config-crypto-map)#set transform-set VPN-SET
Router(config-crypto-map)#match address 100
Router(config-crypto-map)#exit
Router(config)#interface GigabitEthernet 0/1
Router(config-if)#crypto map VPN-MAP
*Jan 3 07:16:24.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Router(config-if)#exit
Router(config)#end
Router#
%SYS-6-CONFIG_I: Configured from console by console
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Router1
Physical Config CLI

IOS Command Line Interface

Router#enable
Router#configure terminal
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#hash sha
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 5
Router(config-isakmp)#exit
Router(config)#crypto isakmp key cisco123 address 10.0.0.1
Router(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
Router(config)#access-list 100 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
Router(config)#crypto map VPN-MAP 10 ipsec-isakmp
Router(config-crypto-map)#set peer 10.0.0.1
Router(config-crypto-map)#set transform-set VPN-SET
Router(config-crypto-map)#match address 100
Router(config-crypto-map)#exit
Router(config)#interface GigabitEthernet 0/1
Router(config-if)#crypto map VPN-MAP
*Jan 3 07:16:24.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Router(config-if)#exit
Router(config)#end
Router#
%SYS-6-CONFIG_I: Configured from console by console

Router#show crypto ipsec sa

Interface: GigabitEthernet0/1
Crypto map tag: VPN-MAP, local addr 10.0.0.2

protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.2.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
current_peer 10.0.0.1 port 800
PERMIT, flags=(origin_is_acl,)
#pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
#pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0

local crypto endpt.: 10.0.0.2, remote crypto endpt.: 10.0.0.1
path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/1
current outbound spi: 0x0(0)

inbound esp sas:
inbound ah sas:

Router#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst src state conn-id slot status

IPv6 Crypto ISAKMP SA

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
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```

```

Router1
Physical Config CLI
IOS Command Line Interface

inbound esp sas:
inbound ah sas:

Router#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst      src      state      conn-id slot status
IPv6 Crypto ISAKMP SA

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#hash sha
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 5
Router(config-isakmp)#exit
Router(config)#
Router(config)#crypto isakmp key cisco123 address 10.0.0.1
A pre-shared key for address mask 10.0.0.1 255.255.255.255 already exists!
Router(config)#end
Router#
%SYS-1-CONFIG_I: Configured from console by console

Router#

Router con0 is now available

Press RETURN to get started.
    
```

PC 0 -

```

PC0
Physical Config Desktop Custom Interface

Command Prompt

Packet Tracer PC Command Line 1.0
PC>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.10: bytes=32 time=0ms TTL=126
Reply from 192.168.2.10: bytes=32 time=0ms TTL=126
Reply from 192.168.2.10: bytes=32 time=0ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

PC>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

PC>
PC>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=0ms TTL=126

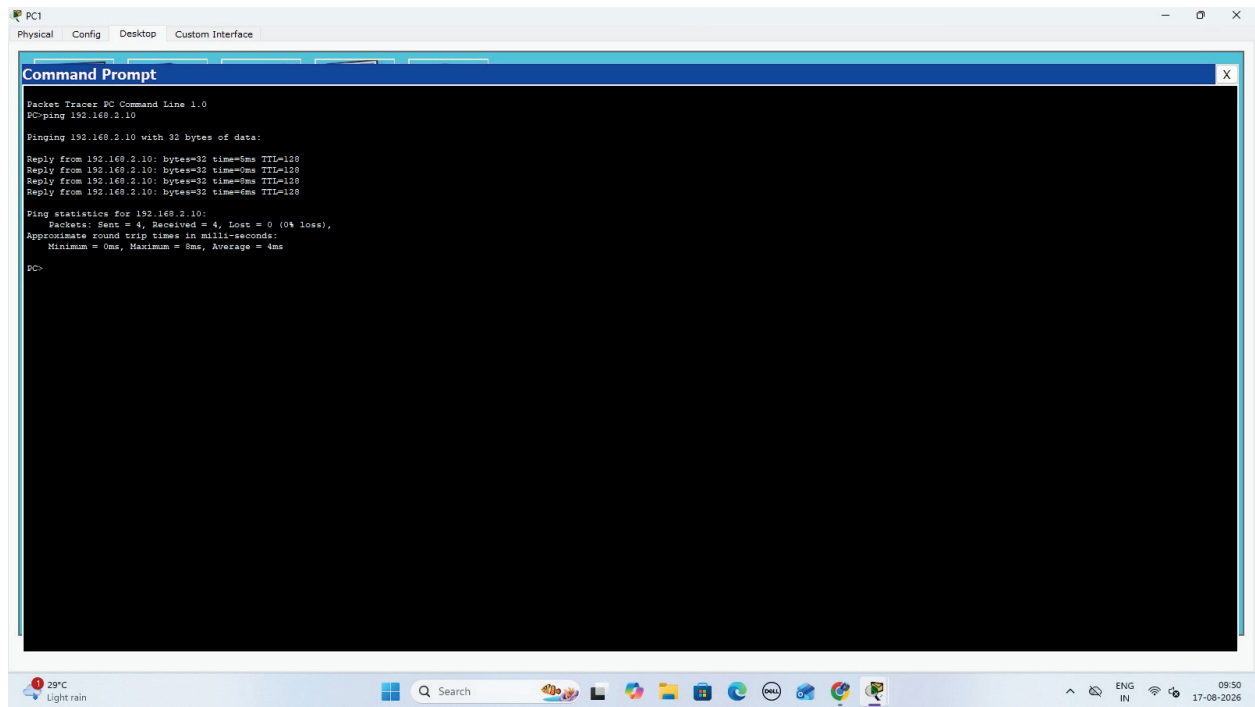
Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

PC>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time=0ms TTL=126
Reply from 192.168.2.10: bytes=32 time=0ms TTL=126
Reply from 192.168.2.10: bytes=32 time=0ms TTL=126
Reply from 192.168.2.10: bytes=32 time=0ms TTL=126
    
```

PC 1 -



The screenshot shows a Packet Tracer PC Command Line window for PC1. The window has tabs for Physical, Config, Desktop, and Custom Interface. The Command Prompt is open, displaying the following text:

```
Packet Tracer PC Command Line 1.0
PC>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time=0ms TTL=128
Reply from 192.168.2.10: bytes=32 time=0ms TTL=128
Reply from 192.168.2.10: bytes=32 time=0ms TTL=128
Reply from 192.168.2.10: bytes=32 time=0ms TTL=128

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 8ms, Average = 4ms
PC>
```

The Windows taskbar at the bottom shows the date and time as 09:50 on 17-08-2026, along with system icons for network, volume, and battery.